MIL-B-43927(GL)

14 April 1975

MILITARY SPECIFICATION

BRIGHTENER, OPTICAL, LAUNDRY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an optical brightener (fluorescent dye) suitable for use with sodium silcofluoride (sour) in the laundering of textile garments (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

P-D-245 - Detergent (Powder, Flake, Bead, or Granular) for Laundry and Dishwashing

PPP-P-45 - Packing of Bulk Quantities of Soap, Detergents, and Related Products (Chip, Powdered or Granular Form) for Domestic and Overseas Shipment

STANDARDS

FEDERAL

FED-STD-536 - Soap and Soap Products (Including Synthetic Detergents) Sampling and Testing

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

FSC 7930

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(Copies of specifications, standards and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The optical brightener shall consist of 25 to 40 percent fluorescent dye or dyes blended with sodium chloride to form a uniform powder. The end product shall not contain any toxic material such as compounds of arsenic, mercury, lead or antimony nor any other material which would have a deleterious effect on the fabric or the user.

3.1.1 Solubility. The optical brightener shall be readily soluable in water at 100°F and (37.8°C) when tested as specified in table I.

3.1.2 Odor. The optical brightener shall have no offensive odor nor shall the product leave any offensive odor on the treated cloth. An offensive odor is defined as rancid, fishy or horsey.

3.1.3 pH value. A one percent distilled water solution of the optical brightener shall have a pH between 8.0 and 12.0 when tested as specified in table I.

3.1.4 Color. Unless otherwise specified, the optical brightener shall have a uniform light color no darker than color number 23722 of FED-STD-595.

3.1.5 Fineness. A minimum of 99.9 percent of the optical brightener shall pass through a No. 50 sieve, when tested as specified in table I.

3.2 Performance. Four specimens of white sheeting shall be laundered in a 0.25 percent solution of detergent, and four specimens of white sheeting shall be laundered in a 0.25 percent solution of the detergent in conjunction with the optical brightener as specified in table I. The whiteness of the specimens washed with detergent and optical brightener shall be a minimum of 30 hue units when compared with the whiteness of test specimens washed in detergent without optical brightener.

3.3 Workmanship. The finished product shall be clean, uniform and free flowing and shall conform to the levels of quality established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise

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specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.1.4. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled shipping containers of the same capacity, manufactured in one batch, for examination in 4.3.1.1 and 4.3.1.2, and in units of shipping containers for examination in 4.3.1.3.

4.3.1.1 Examination of contents of the shipping container. The sample unit for this examination shall be one filled shipping container.

Examine	Defect
Appearance	Not clean; foreign matter present. Not uniform. Not homogeneous.
Color	Darker than color No. 23722 of FED-STD-595.
Odor	Offensive.

4.3.1.2 Examination of the shipping container for net contents. The sample unit for this examination shall be one filled shipping container. The lot shall be unacceptable if the average net content is less than specified.

4.3.1.3 Examination of preparation for delivery requirements. An examination shall be made to determine that preparation for delivery as specified in section 5 is in conformance with PPP-P-45.

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4.3.1.4 Inspection levels and acceptable quality levels (AQLs) for examination. The inspection levels, for determining the sample size, and the acceptable quality levels (AQLs), expressed in defects per one hundred units, shall be as follows:

Examination paragraph	Inspection level	AQLs
4.3.1.1	S-2	2.5
4.3.1.2	S-2	N.A.
4.3.1.3	S-1	2.5

4.3.2 Testing of the end item. The optical brightener shall be tested for the characteristics specified in table I. The sample unit shall be a one pound (0.45 kg) composite. The lot shall be unacceptable if any sample unit fails to meet any test requirement specified. The number of containers to be sampled for obtaining the composite sample, shall be as indicated below:

Lot size (pounds or gallons)	No. of containers
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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INSTRUCTIONS FOR TESTING THE END ITEM

TABLE I

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations Per Unit	Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Comp	Sample		Pass or Fail	Numerically to Nearest		
Composition	3.1	1/	-		-	-			
Solubility	3.1.1	4.4.1	X		1	X			
pH value	3.1.3	4.4.2	X		1	X			
Fineness	3.1.5	4.4.3	X		1	X			
Fabric whiteness (Performance test)	3.2	4.4.4	-		4		1.0 Hue Unit		

1/ The manufacturer shall furnish a certificate of compliance stating that the material composition is in accordance with paragraph 3.1.

AMXRE Form 598
8 February 1965

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4.4 Test methods.

4.4.1 Solubility. Pour 500 ml of warm water (100 ± 5°F) (37.8 ± 1.5°C) into a one liter beaker. Add 5.0 grams of the optical brightener to the water and agitate with a stirring rod for one minute. The optical brightener shall completely dissolve and there shall be no deposit left in the bottom of the beaker. (Do not let stand for a long period of time)

4.4.2 pH of aqueous solution. Dissolve one gram of optical brightener in 100 ml of freshly boiled and cooled distilled water and stir until completely dissolved. Determine the pH value of this solution to the nearest 0.1 unit at room temperature (70 ± 2°F) (21.1 ± 0.6°C) by means of a suitable pH meter which employs the glass electrode.

4.4.3 Fineness. The fineness determination shall be made in accordance with method 2101 of FED-STD-536.

4.4.4 Fabric whiteness (Performance test).

4.4.4.1 Equipment and materials required.

a. Terg-O-Tometer (see 6.3).
b. Gardner XL-10 Colorimeter (see 6.4).
c. Eight swatches of white cotton sheeting measuring 5 by 8 inches (12.7 by 20.3 cm).
d. Detergent conforming to type I of Federal Specification P-D-245.
e. Optical brightener.

4.4.4.2 Procedure.

a. Prepare 4 liters of 0.25 percent detergent solution and heat to 140 ± 2°F (60 ± 0.6°C). Detergent shall be as specified in 4.4.4.1d.
b. Pour 1 liter of the detergent solution in each of the four washing units. Place one test specimen in each of the washing units and launder for ten minutes, maintaining the temperature at 140 ± 2°F (60 ± 0.6°C).
c. Remove swatches, hand squeeze and rinse for five minutes in the Terg-O-Tometer at 140 ± 2°F (60 ± 0.6°C).
d. Press and dry each test specimen by ironing.
e. Repeat operations a thru d except that 0.4 grams of the optical brightener shall be added to the 4 liters of 0.25 percent detergent solution.

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f. Test specimens washed with detergent only (operations a thru d) shall be compared with specimens washed with detergent and optical brightener (operation e) using the Gardner XL-10 Colorimeter. The whiteness of the test specimens washed in the detergent with the optical brightener shall be a minimum of 30 hue units* over the standard when compared with the whiteness of test specimens washed in detergent solution without optical brightener. The fabric whiteness shall be determined using the following formula:

$$W = L + 3a - 3b$$

Where

W = *Fabric whiteness unit
L, a + b = The hue value obtained with normal beam in place

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A, B, and C. Fifty or 100 pounds (22.7 or 45.4 kg) of optical brightener shall be packed in a fiber drum described in PPP-P-45 for level A, B, or C as specified (see 6.2).

5.2 Marking. Marking shall be in accordance with PPP-P-45.

6. NOTES

6.1 Intended use. This item is intended to be used with laundry sour for whitening effects on textiles made from blends of cellulosic and synthetic fibers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Quantity of optical brightener required in shipping containers (see 5.1.1).
(c) Level of packing required (see 5.1).

6.3 Terg-O-Tometer. The Terg-O-Tometer may be obtained from the United States Testing Company, 1415 Park Avenue, Hoboken, NJ 07030 (see 4.4.4).

6.4 Gardner XL-10 Colorimeter. The Gardner XL-10 Colorimeter may be obtained from the Gardner Laboratory, Landy Lane, Bethesda, MD 20014 (see 4.4.4).

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Custodian:

Army - GL

Review activity:

Army - MD

Preparing activity:

Army - GL

Project No. 7930-A309

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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